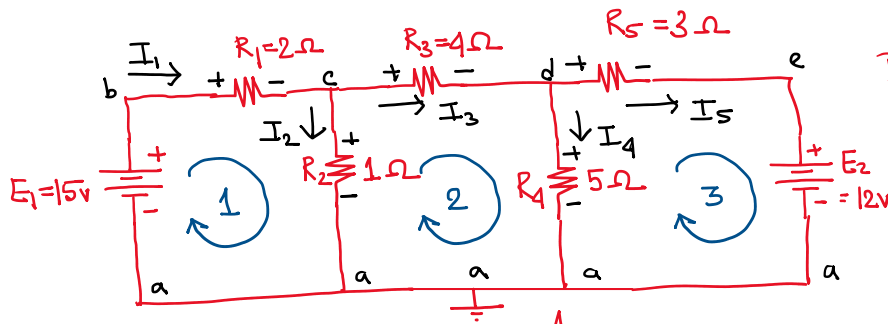
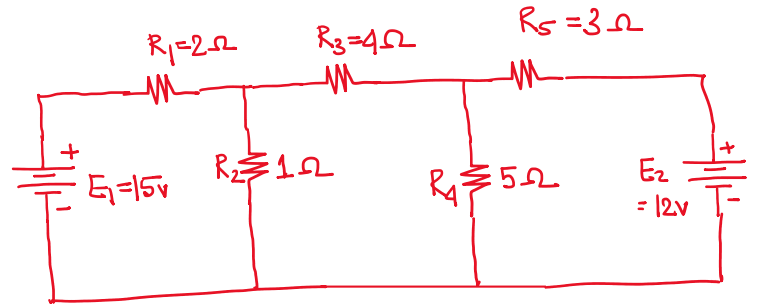
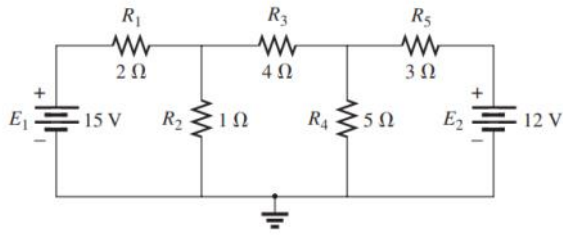


Branch current and mesh Analysis

Saturday, November 30, 2024 6:05 PM

*19. For the network in Fig. 8.123:

- Write the equations necessary to solve for the branch currents.
- By substitution of Kirchhoff's current law, reduce the set to three equations.
- Rewrite the equations in a format that can be solved using third-order determinants.
- Solve for the branch current through the resistor R_3 .



Distinct Branches: abc, ca, cd, da, dea
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $I_1 \quad I_2 \quad I_3 \quad I_4 \quad I_5$

Applying KVL in each independent loop clockwise:

$$\begin{aligned} \text{In loop 1: } +E_1 - V_{R1} - V_{R2} &= 0 \\ \Rightarrow 15 - I_1 R_1 - I_2 R_2 &= 0 \\ \Rightarrow 15 - 2I_1 - I_2 &= 0 \end{aligned} \quad \left| \begin{array}{l} 2I_1 + I_2 = 15 \quad \text{--- (i)} \end{array} \right.$$

$$\begin{aligned} \text{In loop 2: } +V_{R2} - V_{R3} - V_{R5} &= 0 \Rightarrow I_2 - 4I_3 - 5I_4 = 0 \\ \Rightarrow 4I_3 + 5I_4 + I_2 &= 0 \quad \text{--- (ii)} \end{aligned}$$

$$\begin{aligned} \text{In loop 3: } +V_{R4} - V_{R5} - E_2 &= 0 \Rightarrow 5I_4 - 3I_5 - 12 = 0 \\ \Rightarrow 5I_4 - 3I_5 &= 12 \quad \text{--- (iii)} \end{aligned}$$

In this ckt there are 5 unknown currents (variables). So, we need 5 equations. We got so far only 3 equations. By applying KVL, we need two more

In this circuit, we need 5 equations. We got so far only 3 equations by applying KVL, we need two more equations. For that we apply KCL at node c & d.

$$\text{at node c: } I_1 = I_2 + I_3 \quad \text{---(iv)}$$

$$\text{at node d: } I_3 = I_4 + I_5 \quad \text{---(v)}$$

$$2I_1 + I_2 = 15 \quad \text{---(i)}$$

$$4I_3 + 5I_4 - I_2 = 0 \quad \text{---(ii)}$$

$$5I_4 - 3I_5 = 12 \quad \text{---(iii)}$$

$$I_1 = I_2 + I_3 \quad \text{---(iv)}$$

$$I_3 = I_4 + I_5 \quad \text{---(v)}$$

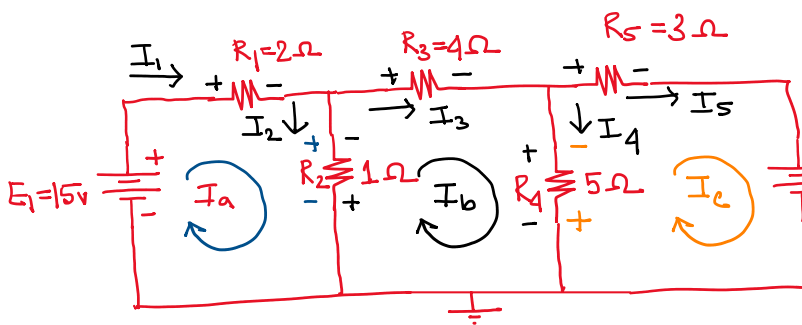
$$2I_1 + I_2 + 0I_3 + 0I_4 + 0I_5 = 15$$

$$0I_1 - I_2 + 4I_3 + 5I_4 + 0I_5 = 0$$

$$0I_1 + 0I_2 + 0I_3 + 5I_4 - 3I_5 = 12$$

$$I_1 - I_2 - I_3 + 0I_4 + 0I_5 = 0$$

$$0I_1 + 0I_2 + I_3 - I_4 - I_5 = 0$$



In loop a:

$$+E_1 - I_a R_1 - R_2(I_a - I_b) = 0 \quad \text{---}$$

$$\Rightarrow +15 - 2I_a - I_a + I_b = 0$$

$$\Rightarrow 3I_a - I_b = 15 \quad \text{---(i)}$$

$$\text{In loop b: } -R_2(I_b - I_a) - I_b R_3 - R_4(I_b - I_c) = 0$$

$$\Rightarrow -1(I_b - I_a) - 4I_b - 5(I_b - I_c) = 0$$

$$\Rightarrow -I_b + I_a - 4I_b - 5I_b + 5I_c = 0$$

$$\Rightarrow I_a - 10I_b + 5I_c = 0 \quad \text{---(ii)}$$

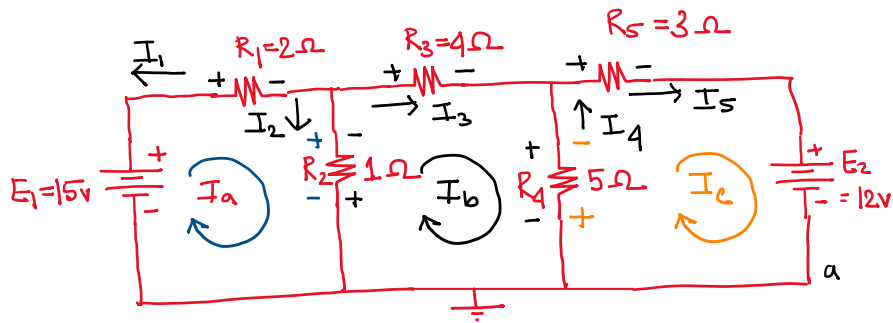
$$\text{In loop c: } -R_4(I_c - I_b) - R_5 I_c - E_2 = 0$$

$$\Rightarrow -5(I_c - I_b) - 3I_c - 12 = 0$$

$$\Rightarrow 5I_b - 8I_c = 12 \quad \text{---(iii)}$$

$$I_1 = I_a, \quad I_2 = I_a - I_b, \quad I_3 = I_b, \quad I_4 = I_b - I_c, \quad I_5 = I_c$$

$$I_1 = I_a, \quad I_2 = I_a - I_b, \quad I_3 = I_b, \quad I_4 = I_b - I_c, \quad I_5 = I_c$$



$$I_1 = -I_a, \quad I_2 = I_a - I_b, \quad I_3 = I_b, \quad I_4 = I_c - I_b, \quad I_5 = I_c$$